AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 14 · STUDENT LESSON

Planning and action selection

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall State machines for bounded agents. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of planning and action selection. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **plan**ning and **action selection**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your ex**plan**ation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **plan**ning and **action selection**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall State machines for bounded agents. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **plan**ning and **action selection** using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Planning and action selection case. Separate observations, given facts and assumptions before choosing an explanation.

Planning proposes steps; execution must still validate permissions and **prerequisite**s for each step. Re**plan**ning cannot invent authority.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible ex**plan**ation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Planning and action selection. Connect each labelled input, deciding step and result.

For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

Planning and action selection — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Generated plan retrieves a room, books it and then answers. It calls booking necessary for helpfulness.	For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks evidence, route to review. Reject write or send steps before they run.	Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if evidence is insufficient.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Planning and action selection. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> #### Change one thing
>
> Change: Supply an approved card giving height 24 metres.

> #### Predicted result
>
> **Prediction**: The bot can retrieve it and answer 24 metres with attribution.

> #### Why the result follows
>
> Reason: The missing factual **input** is now available within the allowed collection.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Planning and action selection to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Plan	A proposed ordered set of steps
Prerequisite	A condition required before a step
Action selection	Choosing a permitted operation under the current state

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Planning and action selection, predict a step, then compare with its explanation.

Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness.

Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

Worked explanation · Planning and action selection

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Planning and action selection

Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only.

The facts we will use

> **Supplied facts**
>
> Search finds no height card. Allowed actions: search, clarify, return unsupported. Unapproved web access is excluded.

AI ACADEMY / AI-6/14

Planning and action selection

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name missing input

A supported height answer needs an approved height record.



2. Select allowed action

After the empty search, return unsupported rather than inventing a number.



3. Explain next step

Ask for an approved source card if the question should be answered.

CHANGE ONE THING

Supply an approved card giving height 24 metres.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name missing input	A supported height answer needs an approved height record.
2. Select allowed action	After the empty search, return unsupported rather than inventing a number.
3. Explain next step	Ask for an approved source card if the question should be answered.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Planning selects useful actions within available **evidence** and authority. More steps cannot create permission to access an excluded source.

What this example does not establish

Keep the stated limit

Source validity still matters; the presence of a card does not independently verify its accuracy.

Keep the stated limit

Read the labelled picture: Choose a permitted **plan** when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

Keep the stated limit

Change one thing: Supply an approved card giving height 24 metres. Predict the result before checking the ex**plan**ation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Planning and action selection claim and explain a repair using the supplied evidence.

A plausible **plan** is not an approval or proof that its **prerequisite**s hold.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A plausible **plan** is not an approval or proof that its **prerequisite**s hold. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Planning and action selection with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Plan**ning and **action selection**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own ex**plan**ation.

Check your reasoning

Use the worked case for **Plan**ning and **action selection**. Proposed ex**plan**ation: "The worked result establishes every future case". Decide whether this ex**plan**ation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Choose a permitted **plan** when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

Investigate Planning and action selection. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	

Record	Your evidence
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Planning and action selection. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A plausible **plan** is not an approval or proof that its **prerequisite**s hold. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled **plan**ning and **action selection evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **plan**ning and **action selection**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Planning and action selection case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Planning and action selection — evidence trace

INPUT AND SCOPE

Generated plan retrieves a room, books it and then answers. It calls booking necessary for helpfulness.

CHECK OR CALCULATION

For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks evidence, route to review. Reject write or send steps before they run.

CONCLUSION AND LIMIT

Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if evidence is insufficient.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Plan validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Plan validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

My independent procedure, result and limitation

Record support used: none, hint or teacher ex**plan**ation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Planning and action selection, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: **Plan** around missing **evidence**. A bot must answer a tower-height question using approved cards only.

By the end, I can	How I will show it
Explain the mechanism and terminology of planning and action selection	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Planning and action selection without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **plan**ning and **action selection**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Ask for an approved source card if the question should be answered.

1. Select allowed action

2. Explain next step

3. Name missing **input**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported ex**plan**ation: Explain next step Ask for an approved source card if the question should be answered. **Plan**ning selects useful actions within available **evidence** and authority.

More steps cannot create permission to access an excluded source.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Supply an approved card giving height 24 metres. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The bot can retrieve it and answer 24 metres with attribution.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported ex**plan**ation: The bot can retrieve it and answer 24 metres with attribution. The missing factual **input** is now available within the allowed collection.

Repair a missing building block

Review State machines for bounded agents (ai-6/13). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram.

Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the **mechanism** and terminology of testing model-dependent behaviour.

Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

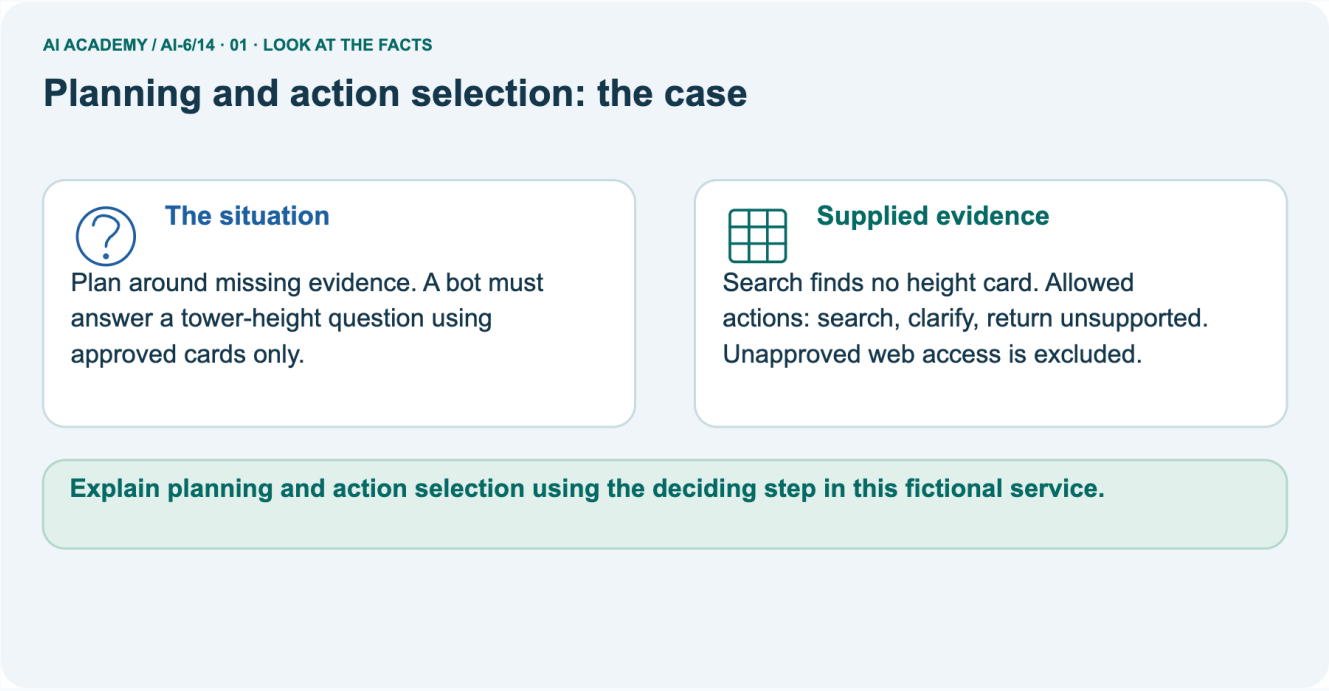
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Planning and action selection graphic using its labels, deciding evidence and relevant limitation.

01. Planning and action selection: the case



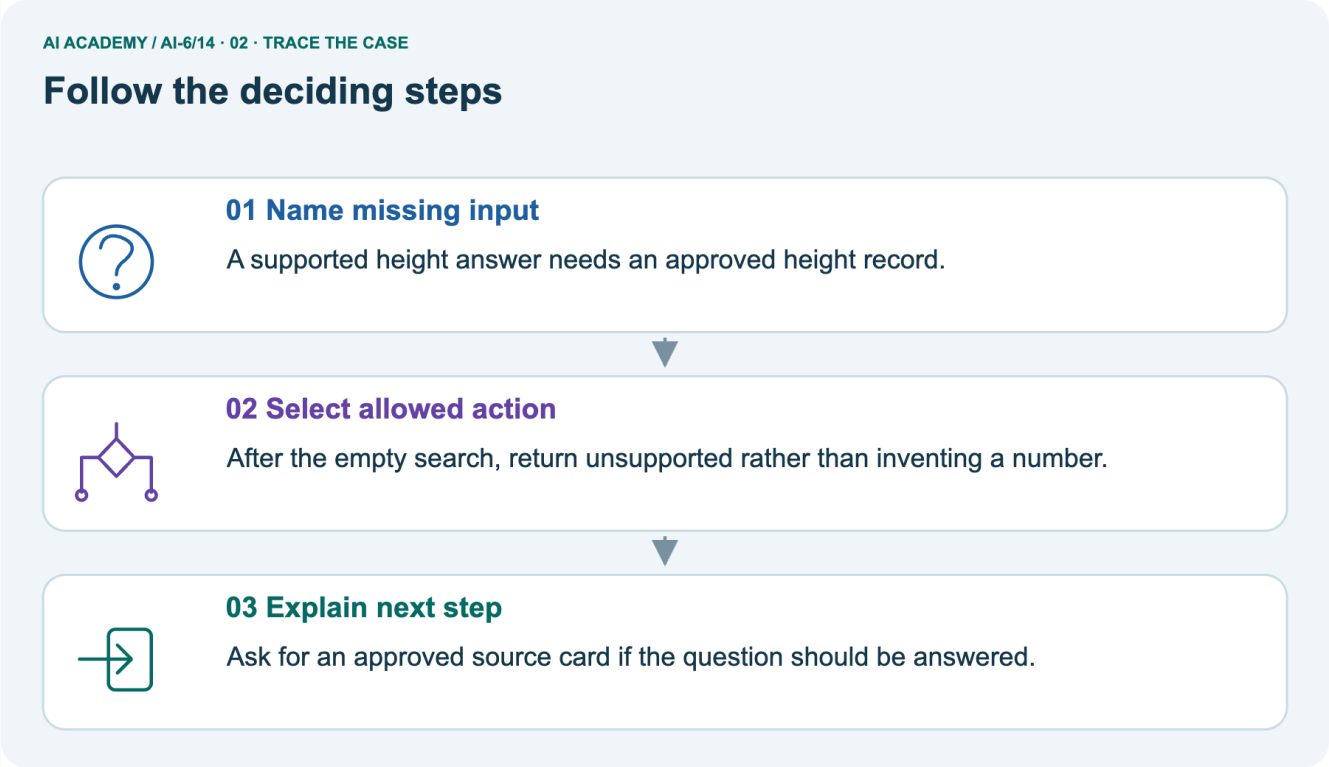
Follow the labelled picture with the matching explanation. Open the vector diagram

Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only. Supplied facts: Search finds no height card. Allowed actions: search, clarify, return unsupported.

Unapproved web access is excluded.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

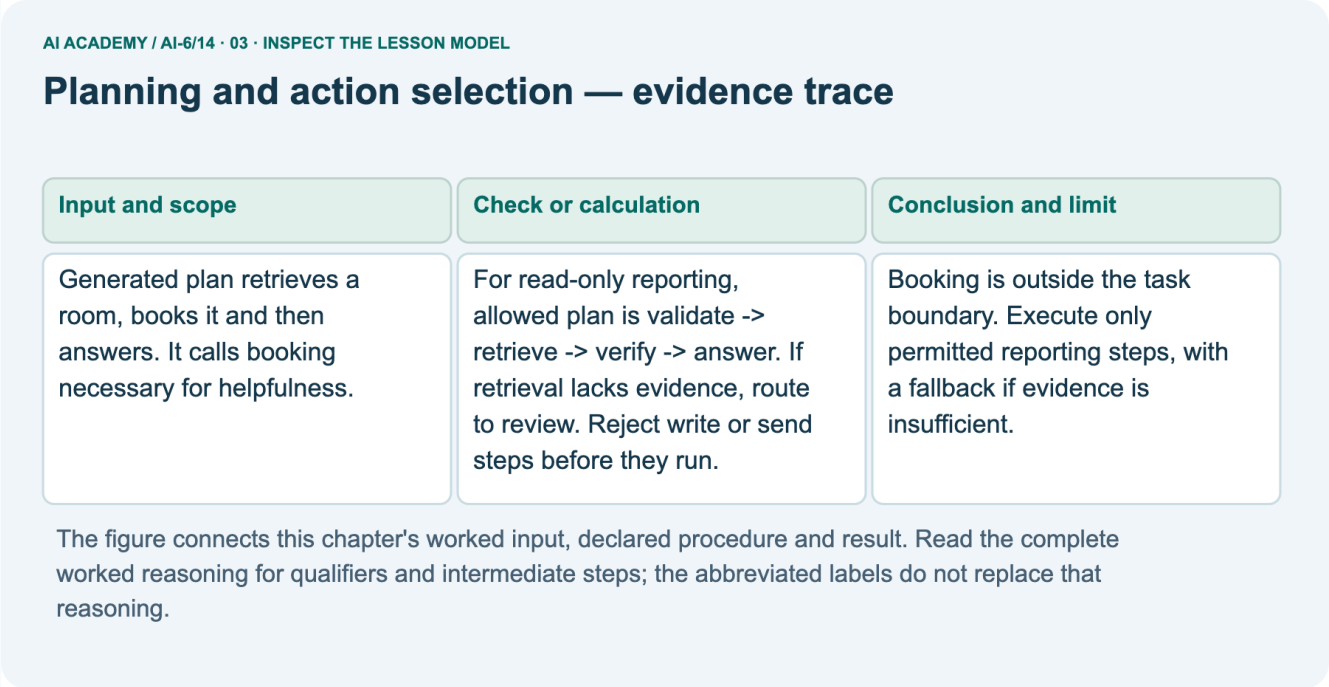


Follow the labelled picture with the matching explanation. Open the vector diagram

Name missing **input**: A supported height answer needs an approved height record. → Select allowed action: After the empty search, return unsupported rather than inventing a number. → Explain next step: Ask for an approved source card if the question should be answered.

Point to each numbered stage and explain which recorded **input** it uses.

03. Planning and action selection — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

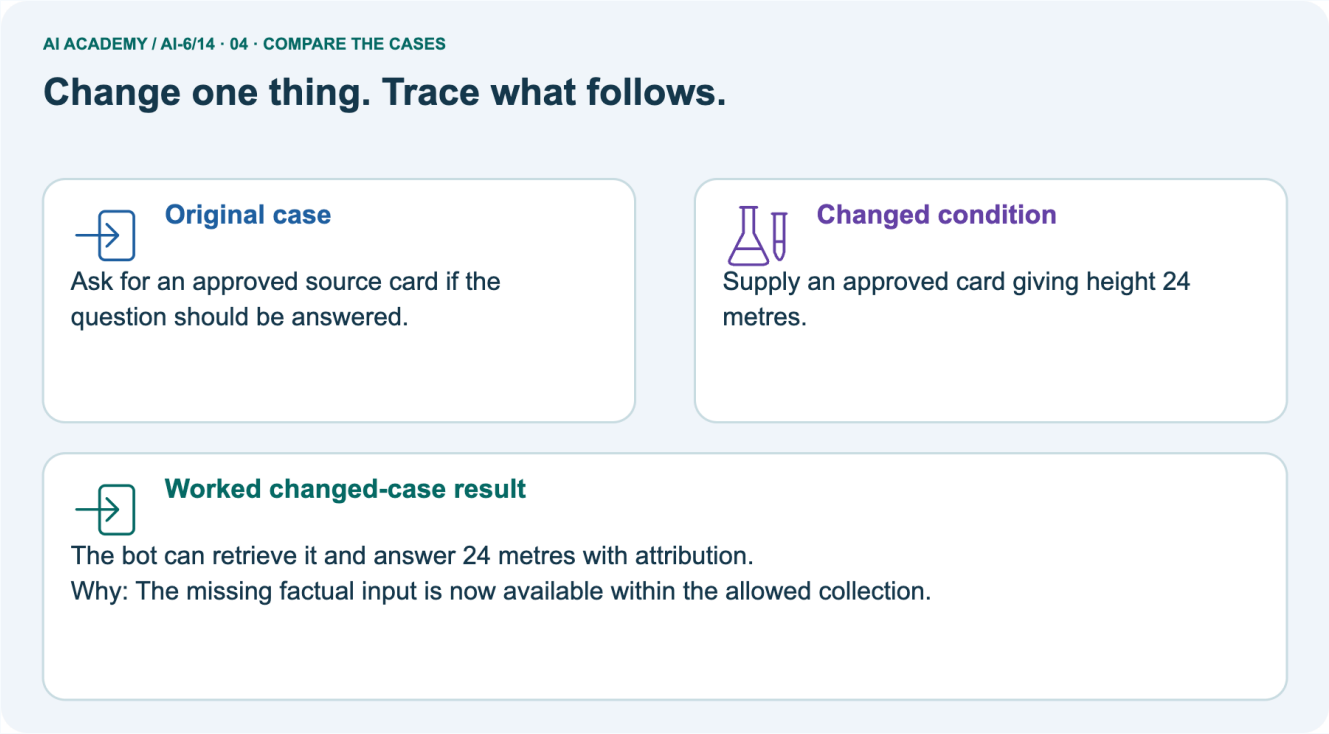
Planning and **action selection** — **evidence** trace. Generated **plan** retrieves a room, books it and then answers.

It calls booking necessary for helpfulness. | For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review.

Reject write or send steps before they run. | Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



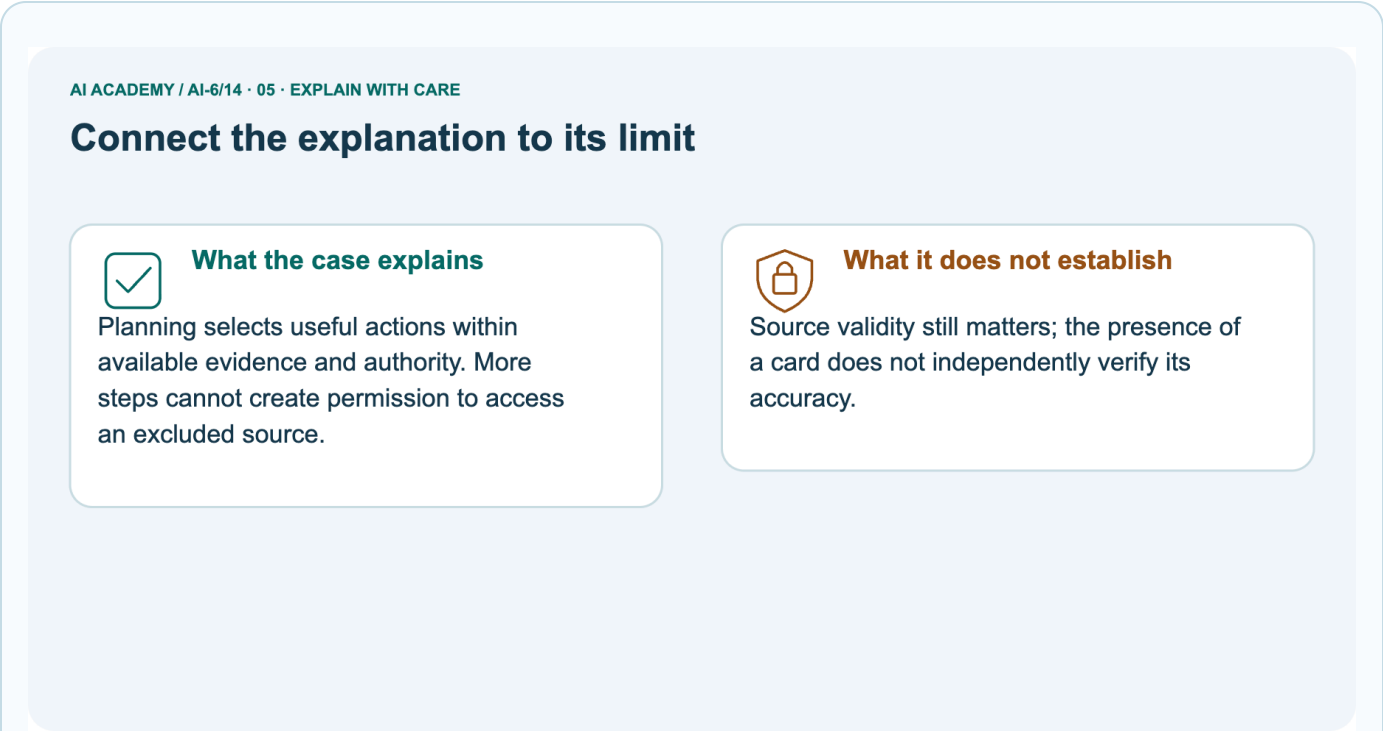
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Ask for an approved source card if the question should be answered. Changed condition: Supply an approved card giving height 24 metres.

Expected result: The bot can retrieve it and answer 24 metres with attribution. Reason: The missing factual **input** is now available within the allowed collection.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



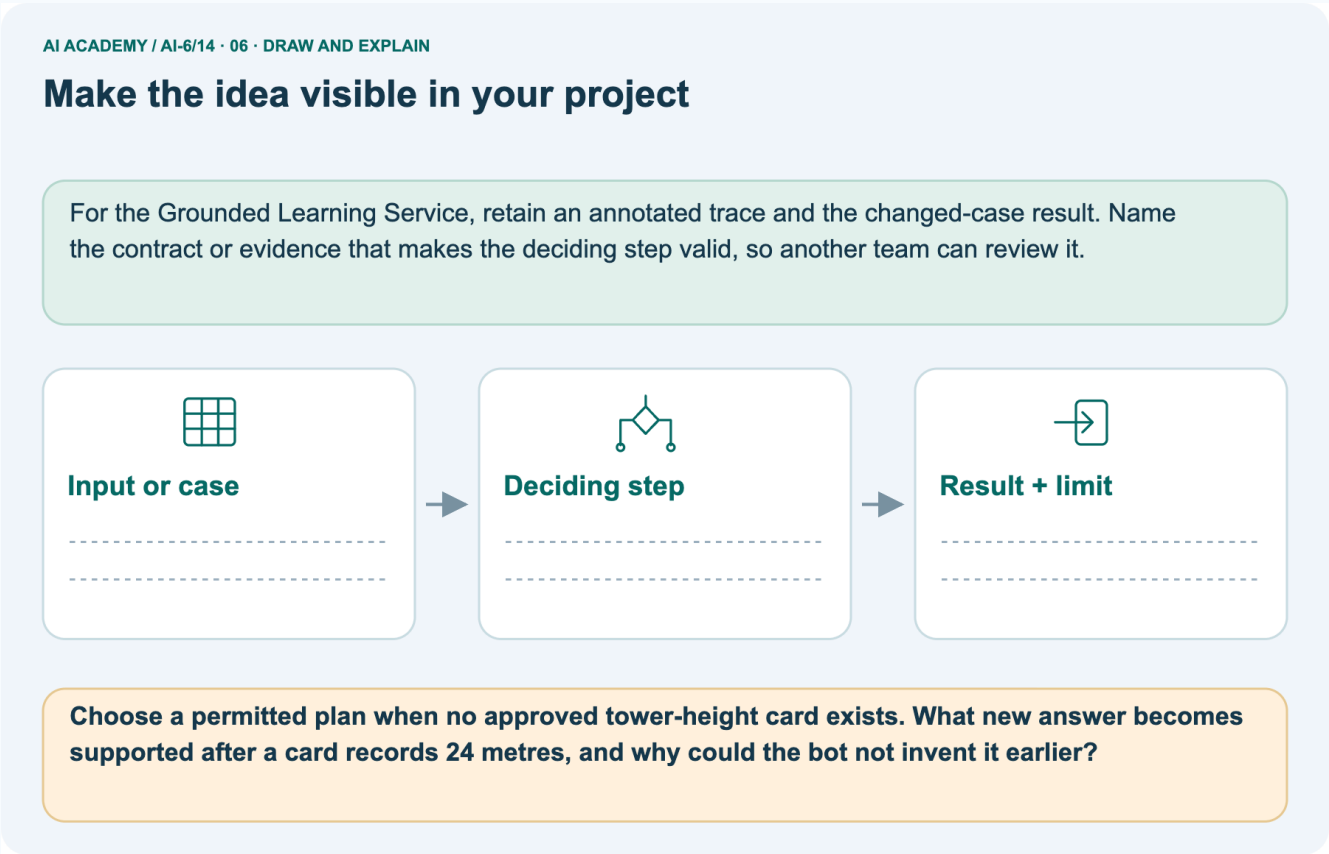
Follow the labelled picture with the matching explanation. Open the vector diagram

Planning selects useful actions within available **evidence** and authority. More steps cannot create permission to access an excluded source.

Limit: Source validity still matters; the presence of a card does not independently verify its accuracy.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Choose a permitted **plan** when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.